

GPBiCG(*m, ℓ, k*)

An Extension of GPBiCG(*m, ℓ*)
and Preliminary Adaptive Parameter Strategies

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I	Krylov reminder	II	GPBiCG(<i>m, ℓ, k</i>)	III	Numerical study	IV	Adaptive strategy	V	Conclusions
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I. Krylov motivation and contribution

Motivation. Product-type Krylov methods avoid long Arnoldi storage, but convergence remains matrix- and parameter-dependent.

Extension. GPBiCG(*m, ℓ, k*) inserts a LOPBiCG-like phase into each GPBiCG(*m, ℓ*) cycle:

$$\text{BiCGSTAB}^m \rightarrow \text{GPBiCG}^{\ell} \rightarrow \text{LOPBiCG}^k.$$

Cycle. Execute (*m*), (*ℓ*), then (*k*) updates in that order. **Tension.** The added phase can accelerate convergence but makes the triplet $\theta = (m, \ell, k)$ harder to choose.

C1. New family
three phases: *m, ℓ, k*

C2. Fair tests
same matrices and baselines

C3. Adaptive angle
rescue then stress test

II. GPBiCG(*m, ℓ, k*) and experimental protocol

Linear systems $b = A(1, \dots, 1)^T, x_0 = 0, r_0^* = r_0.$
Tolerance $\|b - Ax\|_2 / \|b\|_2 \leq 10^{-12},$ except rescue at $10^{-10}.$
Grid 343 MLK triplets and 49 ML pairs.
Compared methods MLK, ML, GPBiCG, BiCGSTAB, BiCGSTAB2.

Matrix	<i>N</i>	nnz	$\kappa_2(A)$	Matrix	<i>N</i>	nnz	$\kappa_2(A)$
utm300	300	3,155	1.50×10^6	memplus	17,758	99,147	1.29×10^5
add20	2,395	13,151	1.20×10^4	epb2	25,228	175,027	2.62×10^3
Goodwin_017	3,317	97,773	1.30×10^5	wang4	26,068	177,196	4.03×10^5
poisson3Da	13,514	352,762	1.12×10^3	crashbasis	160,000	1,750,416	—

Static question
Does *k* improve the grid optimum?

Conventional check
Repeat with ILU(o).

Adaptive question
Can live data reduce sensitivity?

III.a Without preconditioning: residual histories

Figure. Eight true-residual histories without preconditioning. Green frames mark MLK gains; the red frame marks the ML exception. MLK is fastest on 6/8 rows and faster than GPBiCG(*m, ℓ*) on 7/8.

III.b CPU and accuracy without preconditioner

Matrix	Time	MLK Resid.	Error	Time	ML Resid.	Error	Time	GPBiCG Resid.	Error	BiCGSTAB	BiCGSTAB2
utm300	0.0032	4.97e-13	8.13e-11	0.0038	8.00e-13	2.69e-10	0.0403 ¹	3.03e-10 ¹	1.51e-8 ¹	0.0048	0.0055
add20	0.0240	8.53e-13	3.92e-10	0.0222	9.64e-13	5.11e-10	0.0273	9.73e-13	1.19e-9	0.0325	0.0436
Goodwin_017	0.0846	6.38e-13	6.81e-11	0.0974	6.75e-13	1.82e-9	0.1534	9.76e-13	3.17e-10	0.1366	0.2137
poisson3Da	0.0819	5.53e-13	8.75e-12	0.0843	9.30e-13	3.43e-11	0.0964	9.03e-13	1.55e-11	0.0890	0.1024
memplus	0.5491	9.72e-13	1.20e-9	0.5769	8.40e-13	7.33e-10	0.6410	9.88e-13	9.04e-10	0.6924	0.7847
epb2	0.2009	9.14e-13	3.95e-11	0.2124	8.80e-13	1.42e-10	0.2322	7.55e-13	1.16e-10	0.2053	0.2518
wang4	0.2009	8.49e-13	7.80e-11	0.2112	7.49e-13	6.88e-11	0.3285	8.29e-13	1.21e-9	0.1997	0.2982
crashbasis	0.9497	7.80e-13	1.08e-11	1.1351	8.36e-13	1.71e-11	1.4206	1.93e-13	3.31e-12	0.9954	1.3791

Table. CPU seconds, final true residual and offline error. Bold entries mark the fastest CPU row.

6/8
fastest rows

7/8
faster than ML

13–16%
selected reductions vs ML

III.c Conventional ILU(o): residual histories

Figure. The same eight matrices with ILU(o). Green frames mark MLK gains; the red frame marks the ML advantage. Gaps narrow, but MLK remains fastest on 4/7 comparable rows.

V. Conclusions, limitations, and outlook

Limits to state clearly

- The grid $\{1, \dots, 7\}^3$ is systematic but finite.
- CPU rankings depend on compiler flags, cache effects, hardware, and residual logging.
- ILU(o) reduces the room for the added *k*-phase to create large gains.
- The current controller is a first-generation live policy.

Next adaptive step

- Replace blind random starts by a short deterministic portfolio warm-up.
- Freeze (*m, ℓ, k*) after stable phase evidence and reactivate only during stagnation.
- Add cheap matrix-family, non-normality, or spectral proxies to guide the first triplet.
- Normalize phase scores by computational cost.

References

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6/8
fastest static
7/8 faster than ML

4/7
fastest with ILU(o)
narrower gains

2744
paired starts
selected rescue > 2×

800
random starts
1/8 beats best no-tuning

Key message

GPBiCG(*m, ℓ, k*) adds a third product-type phase. Static evidence is strong; live control can rescue poor triplets. Full no-tuning superiority remains a research direction.

III.d Conventional ILU(o): CPU table

Matrix	Best MLK	Best ML	CPU MLK	CPU ML	vs GPBiCG	vs ML
utm300	GPBiCG(7,5,2) [†]	GPBiCG(3,4) [†]	—	—	—	—
add20	GPBiCG(2,3,1)	GPBiCG(2,4)	0.0136	0.0133	1.13	0.982
Goodwin_017	GPBiCG(7,7,3)	GPBiCG(2,5)	0.0647	0.0587	—	0.907
poisson3Da	GPBiCG(6,5,5)	GPBiCG(4,3)	0.0673	0.0710	1.21	1.05
memplus	GPBiCG(3,5,4)	GPBiCG(7,3)	0.1884	0.1973	1.37	1.05
epb2	GPBiCG(3,3,3)	GPBiCG(7,4)	0.0304	0.0305	1.20	1.00
wang4	GPBiCG(4,5,6)	GPBiCG(6,3)	0.0469	0.0497	1.23	1.06
crashbasis	GPBiCG(7,7,2)	GPBiCG(7,7)	0.1949	0.1815	1.21	0.931

Table. Speedups larger than one favor MLK. Dagger rows are diagnostic/non-converged. MLK is fastest on 4/7 comparable rows; gains are narrower after ILU(o).

III.e Static speedup synthesis

Figure. Article CPU speedups. Values above one favor MLK; blank or “n.c.” entries identify non-comparable diagnostic rows where status differs.

IV.a Adaptive controller: one cycle, one decision

1. Observe. Run one complete cycle *q* with fixed $\theta_q = (m_q, \ell_q, k_q) \in \Theta = \{1, \dots, 7\}^3$. For each phase $p \in \{B, G, L\}$, compute its mean contraction per iteration,

$$\rho_{q,p} = \left(\frac{\|r_{q,p}^{\text{out}}\|_2}{\|r_{q,p}^{\text{in}}\|_2} \right)^{1/n_{q,p}}, \quad g_q = \frac{\|r_q^{\text{out}}\|_2}{\|r_q^{\text{in}}\|_2}.$$

2. Propose. Let $p_+ = \arg \min_p \rho_{q,p}$ and $p_- = \arg \max_p \rho_{q,p}$. Transfer one iteration: $\theta' = \theta_q + e_{p_+} - e_{p_-}$, with $e_B = (1, 0, 0)$, $e_G = (0, 1, 0)$, $e_L = (0, 0, 1)$.

3. Accept or keep. The next cycle uses

$$\theta_{q+1} = \begin{cases} \theta', & g_q < 1, \max_p \rho_{q,p} < 1, \theta' \in \Theta, \\ \theta_q, & \text{otherwise.} \end{cases}$$

Cycle *q*
fixed θ_q

→

Measure
 $g_q, \rho_{q,p}$

→

Propose
local θ'

→

Decide
3 tests

→

Cycle *q* + 1
accept / keep

IV.b Experiment I: prescribed-start rescue

CPU ratio = static / live

Question. Can live information rescue an unfavorable prescribed triplet?
Protocol. Pair each static run with a live run from the same initial triplet.

2744
paired starts

69.1%
crashbasis wins

0.968
median ratio

> 2×
selected rescues

Figure. Static/live CPU ratios for all eight matrices and all 343 prescribed starts at 10^{-10} . Ratios above one favor live adaptation.

Reading. Selected poor starts are rescued; the median (0.968) exposes monitoring overhead when rescue is unnecessary.

IV.c Experiment II: fully parameter-free starts

Figure. 100 seeded starts per matrix at 10^{-12} ; median and 10–90% envelope. Red: low convergence. Green: sole gain vs best classical baseline.

88%
convergence / 800 runs

5/8
faster than GPBiCG

1/8
beats best no-tuning

Matrix	Conv.	PF CPU	Baseline	Best/PF
utm300	66%	0.0051	0.0048	0.95
add20	100%	0.0315	0.0273	0.87
Goodwin_017	49%	0.1605	0.1366	0.85
poisson3Da	100%	0.0885	0.0890	1.01
memplus	97%	0.6646	0.6410	0.96
epb2	99%	0.2277	0.2053	0.90
wang4	93%	0.3000	0.1997	0.67
crashbasis	100%	1.1870	0.9954	0.84

Best/PF > 1 favors the parameter-free method; the baseline is the best of GPBiCG, BiCGSTAB, and BiCGSTAB2.

Current limitation. Blind-uniform starts remove manual triplet selection, but do not yet yield a winning no-tuning solver. Monitoring and transitions are included.